

Sirui Zou

+1 510-542-3302 | siruizou2005@gmail.com | github.com/siruizou2005

EDUCATION

- University of California, Berkeley (UC Berkeley)** Jan 2026 – May 2026
Berkeley Global Access (BGA) Program Focus: Economics & Computational Social Science Berkeley, CA, USA
- **Core Courses:** ECON141 (Econometrics), Psych C123 (Computational Models of Cognition), Challenge Lab
- Peking University (National School of Development)** Jul 2025 – Aug 2025
Summer School: Quantitative Finance Beijing, China
- **Grade:** A+ (Top 10/150). Focused on stochastic analysis and quantitative pricing models.
- Southwestern University of Finance and Economics (SWUFE)** Sep 2024 – Jun 2028
Economics Chengdu, China
- **Academics & Quant:** GPA: 3.9/4.0 | GRE Quant: 170/170 | GMAT FE: 655 (Top 9% Globally, Pre-enrollment)
 - **Core Courses:** Macroeconomics (A+), Python (A+), Advanced Math (A), Linear Algebra (A), Probability & Statistics (A-)
 - **Scholarships:** Academic Scholarship ('24, '25), Innovation & Entrepreneurship Scholarship ('24)

ACADEMIC & RESEARCH EXPERIENCE

- PersonaForge: Role-Playing Agent Research based on Dual-Process Architecture** Nov 2025 – Jan 2026
- **Role:** Co-first Author. Submitted to **ACL 2026** (Under Rolling Review)
 - **Method:** Designed a dual-process cognitive architecture fusing "System 1/System 2" and Big Five personality traits to build a dynamic defense mechanism addressing LLM "persona drift" in long-text simulations.
 - **Results:** Reduced persona drift by **75%** and token cost by **87%** (maintaining 96% performance) across 50 dialogue rounds with 88 characters.
- LLM-Assisted Consumption Policy Efficacy Quantification & Evaluation** Jan 2026 – Present
Research Assistant (Advisors: Prof. Hong Zou, Dean of School of Economics, & Dr. Han Xiao)
- **Methodology Design:** Developing an automated NLP pipeline using fine-tuned open-source LLMs (e.g., Qwen) to score and quantify "consumption policy" intensity across an estimated 3–5M government documents (2000–2022).
 - **Evaluation Framework:** Designing a multi-dimensional system to calculate macroeconomic ROI on GDP and retail sales, proposing to integrate CHFS data to measure heterogeneous impacts on household Marginal Propensity to Consume (MPC).

TECHNICAL & ENGINEERING PROJECTS

- ScrollWeaver & Soulverse: Multi-Agent Social Simulation Engine** Team Lead Oct 2025 – Dec 2025
- **Project Overview:** Designed and developed a multi-agent social simulation engine that breathes life into static texts (novels, worldbuilding documents, lore collections), transforming them into interactive, co-creatable, and exportable living worlds. (Global 3rd Prize, 2025 Soul AI Agent Competition)
 - **Technical Implementation:** Built a LangChain-based Orchestrator-Performer architecture with FastAPI/WebSockets for real-time streaming and D3.js for dynamic maps. Empowered multi-LLM agents (GPT, Claude) with autonomous logic and social deduction via RAG, ChromaDB/SQLite memory, a 3-layer personality model, and a dual-process cognitive architecture.
- Course Prism: Course Evaluation & Retrieval Community** Private Developer Mar 2025 – Jun 2025
- **Development:** Independently developed SWUFE's open-source course evaluation platform. Built interactive UI via Next.js/React and implemented RESTful APIs/unified auth using Django REST, PostgreSQL, and Redis.
 - **Deployment:** Configured Docker Compose, Nginx reverse proxy, and Huey async task queues for one-click containerized deployment, automated data backup, and rapid recovery.

SKILLS & INTERESTS

Tools: Python (PyTorch, Pandas, NumPy), SQL, Stata, LaTeX, Linux, Docker, Git

Interests: Computational Social Science, Economics, NLP, LLM Agents

Languages: English (IELTS 7.0), Mandarin (Native)